

CASE REPORT

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Spontaneous regression of large-sized lumbar facet synovial cysts: two case reports and literature review

Hong Jin Kim^{1,2} , Jong-Hyun Ko³ and Dong-Gune Chang^{2,4*}

Abstract

Background Lumbar facet synovial cysts (LFSCs) are an uncommon degenerative lesion in the lumbar spine, but treatment remains controversial. We report two rare cases of spontaneous regression of large-sized LFSCs.

Case presentation A 68-year-old male visited our outpatient clinic with a complaint of lower back pain and right lower extremity radiculopathy beginning approximately two weeks prior. Computed tomography of the spine showed facet arthrosis, and magnetic resonance imaging (MRI) of the spine revealed a large, oval cystic-mass-like lesion at the L5–S1 level. After conservative treatment for one year, the patient's symptoms had completely resolved, and a follow-up MRI showed spontaneous regression of the large LFSC. A 69-year-old female with a complaint of radiating pain on left lower extremity and intermittent claudication that had aggravated about 2 months ago also showed a 1.5 × 1.2 × 1.7 cm oval cystic mass involving left L5, S1 nerve roots at L5–S1 level from the MRI. After conservative treatment for three months, a follow-up MRI also showed spontaneous resolution of the large LFSC.

Conclusions The large-sized LFSCs spontaneously regressed without any surgical intervention. If symptoms can be controlled, conservative treatment should be considered for first-line treatment despite a mismatch between radiological and clinical findings. Careful clinical judgments are essential to prevent unnecessary surgery.

Keywords Synovial cysts, Degeneration, Facet arthrosis, Conservative treatment, Spine

Background

Since 1877, when Baker first described synovial cysts near a degenerative knee joint, lumbar facet synovial cysts (LFSCs) have been reported as an uncommon cause of radicular pain by Vosschulte and Borger in 1950 and Kao et al. in 1974 [1–3]. Generally, LFSCs are considered to be uncommon degenerative lesions in the lumbar spine that usually produce non-specific symptoms such as back pain or radiculopathy [4]. Advances in imaging modalities such as computed tomography (CT) and magnetic resonance imaging (MRI) now enable accurate diagnosis and preoperative planning [2, 4]. Still, treatment protocols, from observation to surgical resection, have not yet been established because LFSCs are rare [2, 5]. With the paucity of literature, clinicians must make their own treatment decisions when faced with a large

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LFSC compressing the thecal sac. Here, we describe two cases in which large-sized LFSCs spontaneously regressed, which provides useful information for therapeutic decision-making.

Case presentation

Case 1

A 68-year-old male visited our outpatient clinic with a complaint of lower back pain and right lower extremity radiculopathy that had begun about two weeks prior. The visual analogue scale (VAS) was 6, indicating mild to severe pain. Physical examination demonstrated positive signs at 50° in a straight leg raising test. Neurological examination did not reveal any specific signs, such as paralysis or sensory decrease. Plain radiographs of the spine showed degenerative spondylolisthesis at L4–5 and the lumbarization of S1 (Figs. 1a and b). CT of the lumbar spine showed severe right-sided facet joint arthrosis, suggesting degenerative changes at the L5 and L5–S1 levels (Fig. 2). MRI of the lumbar spine revealed a 1.8 cm × 1.7 cm × 1.0 cm oval cystic-mass-like lesion at the L5–S1 level (Fig. 3a), with high signal intensity (SI) on T1-weighted imaging (WI) (Fig. 3b) and low SI on T2-WI (Fig. 3c).

Because the line of the cystic mass continued from the right facet joint, the diagnosis of LFSC caused by facet joint arthrosis was made. We recommended surgical excision and decompression for the large-sized oval-shaped cystic mass because the patient complained of back pain and radiculopathy. However, the patient refused surgical treatment, expressing a strong

preference for pain control through medication. Therefore, we opted for conservative treatment with regular follow-up of symptoms and neurological signs in our outpatient clinic. At the three-month follow-up, the VAS score was 2, and the pain was tolerable with medication. The patient's symptoms gradually improved, and no neurological signs were observed. At the one-year follow-up, the patient's symptoms had completely resolved, and the follow-up MRI showed spontaneous regression of the large-sized LFSC (Figs. 3d–f). The patient was still doing well without recurrence at the 2-year follow-up visit.

Case 2

A 69-year-old female visited our hospital outpatient unit with a complaint of radiating pain on left lower extremity and intermittent claudication that had aggravated about 2 months ago. On the physical examination, her myotome of left L5 (grade 5-) and S1 (grade 4) were deterioration without sign or symptom of cauda equina syndrome. There was mild scoliotic deformity but no significant abnormality at L5–S1 level, such like spondylolysis and segmental instability, on plain radiographs (Figs. 4a and b). MRI performed at the time of symptom onset revealed a 1.5 × 1.2 × 1.7 cm oval cystic mass involving left L5, S1 nerve roots at L5–S1 level (Fig. 5a), with high SI on T2-WI (Fig. 5b) and iso-SI on T1 WI (Fig. 5c).

Given that the base of the cystic mass was connected to the left L5–S1 facet joint, a radiologic diagnosis of the LFSC secondary to facet joint arthrosis was also made. We recommended surgical decompression even if the



Fig. 1 Plain radiographs of the first case. **a** and **b**, Degenerative spondylolisthesis at L4–5 and lumbarization of S1

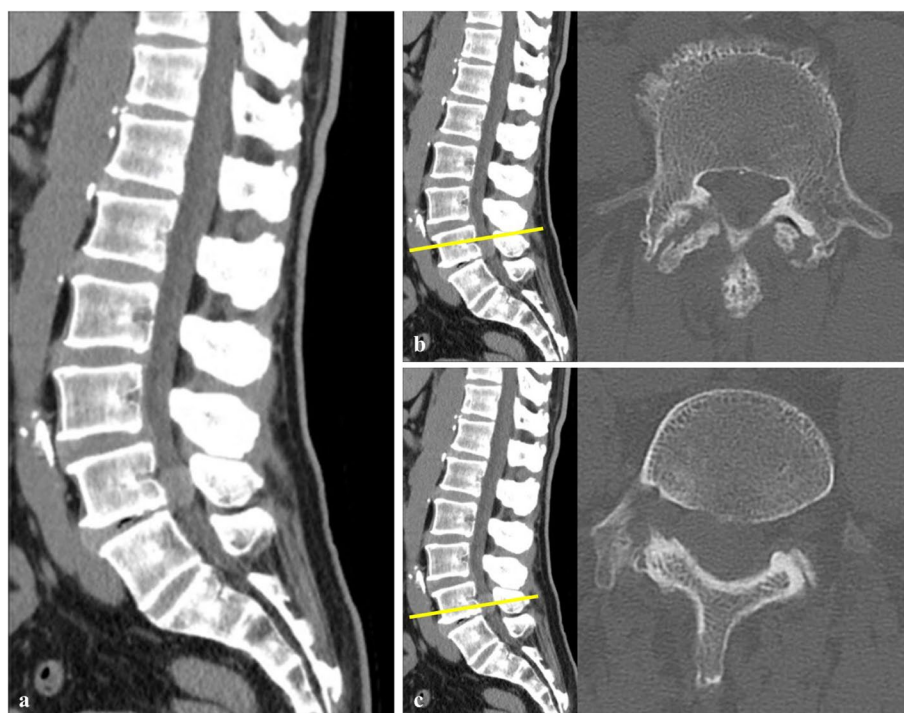


Fig. 2 Computed tomography (CT) of the lumbar spine in the first case. **a** Sagittal view of CT presenting a large, oval abnormal signal in the thecal sac at the L5–S1 level. **b** and **c**, Axial view of CT showing severe right-sided facet joint arthrosis, suggesting degenerative changes at the L5 and L5–S1 levels

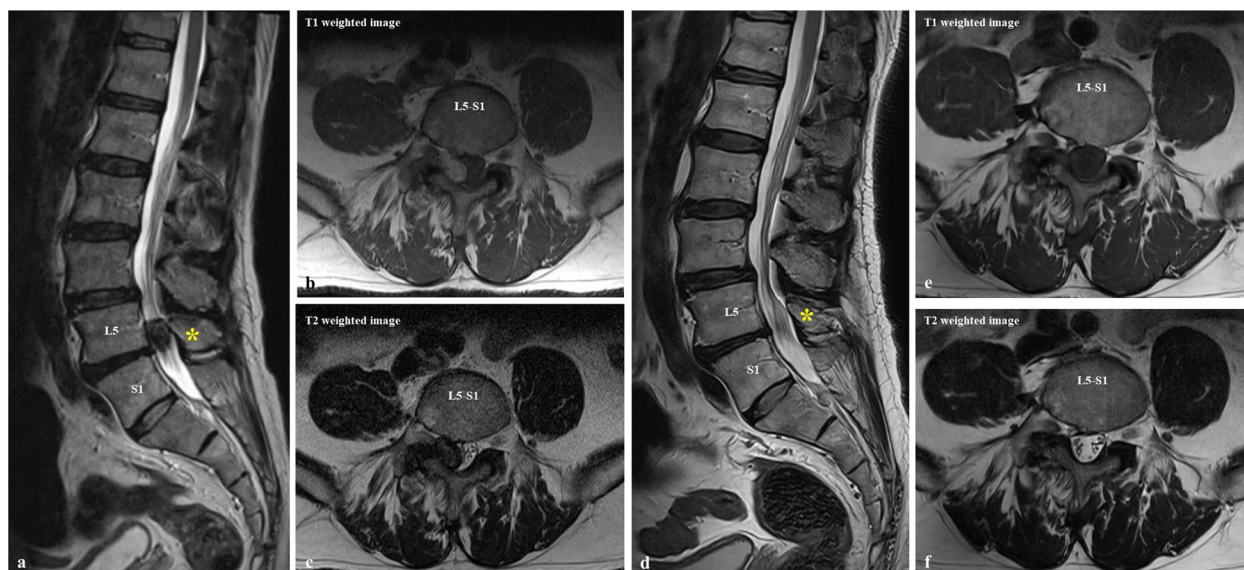


Fig. 3 Magnetic resonance imaging (MRI) of the lumbar spine in the first case. **a** A 1.8 cm × 1.7 cm × 1.0 cm oval cystic-mass-like lesion compressing the thecal sac at the L5–S1 level. **b** and **c**, A large, oval cystic mass presented high signal intensity (SI) on T1-weighted imaging (WI) and low SI on T2-WI. **d**, **e**, and **f**, At the one-year follow-up visit, the patient's symptoms had completely resolved, and the follow-up MRI showed spontaneous regression of the large-sized LFSC



Fig. 4 Plain radiographs of the second case



Fig. 5 Magnetic resonance imaging (MRI) of the lumbar spine in the second case. **a** $1.5 \times 1.2 \times 1.7$ cm oval cystic mass involving left L5, S1 nerve roots at L5–S1 level. **b** and **c**, A large, oval cystic mass presented high signal intensity (SI) on T2-weighted imaging (WI) and iso SI on T1-WI. **d**, **e**, and **f**, At the three-month follow-up visit, the patient's symptoms had completely resolved, and the follow-up MRI showed spontaneous regression of the LFSC

radiating pain and claudication persist after 4 weeks of conservative treatment. However, the patient's symptoms and motor strength improved after four weeks of medication for pain control. We checked the follow-up MRI, which also showed spontaneous regression of the large-sized LFSC (Figs. 5d-f). The patient was still doing well without any complaints at the latest outpatient follow-up.

Discussions

LFSCs are cystic masses that arise from the facet joint (zygapophyseal joint) capsule of the lumbar spine, and they include juxta-facet cysts, both synovial and ganglion [3, 6–8]. As synovial cysts are caused by degeneration, the pathomechanism of LFSCs is based on protrusion of the synovial membrane through joint capsule defects [8].

Repetitive microtrauma, segmental instability, and subsequent degenerative changes lead to facet joint arthrosis, which leads to the occurrence of LFSCs [2, 8]. The most common LFSCs occur at L4–5, which is the most mobile segment of the spine [2]. Based on those pathomechanisms, cyst excision, decompression, and instrumented fusion are considered to be effective surgical treatments that correct the instability and prevent recurrences [5]. However, the surgical treatment of LFSCs remains controversial because of the risk of mechanical complications after instrumented fusion [9–11]. Our patients presented with large-sized LFSCs at L5–S1 but complained of only mild to moderate back pain and radiculopathy, despite thecal sac compression. Moreover, the large-sized LFSC spontaneously regressed, during which symptoms gradually improved. Therefore, our cases suggest that conservative treatment can be considered for facet cysts, even those with thecal sac compression, if symptoms are not severe.

Synovial cysts in the spine are uncommon, with a 0.65 to 2.50% incidence rate in the general population [4, 12]. Bydon et al. reported that 96.2% of LFSCs occur in the lumbar spine: at L4–5 (about 68.0%), L3–4 (14.1%), and L5–S1 (15.2%) [2]. When considering the lumbarization of S1 in our case, the LFSC occurred in the most mobile region (L5–S1). Several previous cases have reported the spontaneous regression of LFSCs with only conservative management, similar to our cases [2, 8]. However, those studies reported that very large cystic masses cannot be successfully addressed with conservative treatment when the thecal sac is compressed, which our case findings contradict. Considering this, our cases are not only rare but also demonstrate a mismatch between radiological findings and clinical symptoms, suggesting the importance of clinical judgment. Campbell et al. classified LFSCs based on canal compromise and listhesis, finding no correlation between canal compromise and clinical presentation [13]. Recurrence following decompressive surgery was more common in cases with listhesis [13]. However, limited evidence exists regarding the relationship between cyst size and its resolution with conservative management. Based on Campbell's findings, we thought that cyst size in our two cases does not directly correlate with symptom severity [13]. Therefore, treatment strategies of LFSCs should be determined by several factors, including instability and symptom severity.

The differential diagnosis for facet joint disorders includes hematoma, benign bone tumors, and degenerative diseases such as disc herniation and synovial cysts. Anaya JEC et al. described that facet cysts are the most prevalent benign cystic lesions, typically presenting high SI on T2-WI and low SI on T1-WI [14]. Although we did not make an exact diagnosis through surgical biopsy, the

degenerative spondylolisthesis at L4–5 and facet arthrosis suggest that our mass was likely an LFSC, not a meningioma or hematoma [6]. Our radiological approach for diagnosing this as a cystic mass was based on finding it to be a homogeneous mass, with high SI on T1-WI and low SI on T2-WI [6]. Meanwhile, the second case showed homogeneous mass, with iso-SI on T1-WI and high SI on T2-WI. Signal intensity of LFSC in MRI is associated with the chronicity and characteristics of a cystic mass [15]. Typically, the cyst appears with high SI on T2-WI; however, low SI on T2 can also be observed if there is a hemorrhagic component or if it reflects the early phase of LFSC [15]. This pattern corresponds to the SI of the facet joint in our cases. Furthermore, the line connecting the cystic mass and the facet joint suggested a mass-like lesion that originated from the facet joint [2, 6, 8]. The facet joint effusion was also detected on the follow-up MRI of our 2nd case, along with a residual cyst communicating with the facet joint. This suggests that the LFSC resolved into the facet joint, leading to the accumulation of fluid within the joint, which was subsequently detected. However, a limitation of our study is that we were unable to observe the progression of LFSC resolution during the interim period. Therefore, careful clinical judgment about whether the cyst is continuous with the facet joint on MRI is needed during treatment planning to prevent unnecessary surgery.

The treatment strategy for LFSCs remains controversial, but it is primarily determined by the patient's symptoms [2]. Asymptomatic synovial cysts should be observed, with no need for specific treatment [8]. As previously mentioned, they can spontaneously resolve over several months [2]. In our cases, spontaneous regression of LFSCs was observed following a successful response to conservative management. However, in cases where conservative management is ineffective, more invasive interventions such as fluoroscopic-guided steroid injection with cystic rupture may be viable treatment options [2]. Martha et al. reported that the lumbar facet steroid injection combined with an attempt at cyst rupture was associated with avoiding subsequent surgery in approximately half of the treated patients [16]. When a cyst is accompanied by intractable pain or neurological signs with no response to conservative treatment or intervention, surgical treatment for cyst excision and hemilaminectomy is often attempted [1, 2]. Instrumented fusion can be used to support the stability of the lumbar spine in addressing pre-existing segmental instability. [4, 5]. Although our case had pre-existing segmental instability with severe facet joint arthrosis, it is noteworthy that the large LFSC spontaneously regressed and had no recurrence thereafter. Therefore, it is crucial to consider conservative treatment for patients with large LFSCs when

their symptoms are under well managed, although close observation is needed due to the potential for neurological signs to occur at any time.

In conclusion, our cases suggest that large LFSCs spontaneously regressed without any surgical intervention. If patient symptoms are controlled, conservative treatment should be considered as first-line treatment despite the mismatch between radiological and clinical findings. Careful clinical judgment is essential to avoid unnecessary surgery. However, further studies, such as case series or case-control designs, are necessary to support our conclusion.

Abbreviations

LFSC	Lumbar facet synovial cysts
CT	Computed tomography
MRI	Magnetic resonance imaging
SI	Signal intensity
WI	Weighted imaging

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None

Authors' contributions

HJK; made substantial contributions to the conception of the work, the acquisition, analysis, and interpretation, and drafted the original manuscript. J-HK; made substantial contributions to manuscript revision and funding acquisition. D-GC; made substantial contributions to the conception of the work and revised the article critically, agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved, approved the paper for publication.

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Data availability

The data that support the findings of this study are not openly available due to reasons of sensitivity and are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

This case report was approved by the Institutional Review Board (IRB) of Inje University Sanggye Paik Hospital (IRB no. 2024-03-001). The authors have obtained written informed consent from the patient to publish this case report and any accompanying images.

Consent for publication

The authors have obtained written informed consent from the patient to publish this case report and any accompanying images.

Competing interests

The authors declare no competing interests.

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